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The Tagore utility for genome feature illustration on human chromosome ideograms

Aroon T. Chande[1*], Lavanya Rishishwar[1,2*], Andrew B. Conley[1,2], King Jordan[1,3], and Leonardo Marino-Ramírez[2¶]

1 IHRC-Georgia Tech Applied Bioinformatics Laboratory, Atlanta, Georgia, USA **2** National Institute on Minority Health and Health Disparities, National Institutes of Health, Bethesda, Maryland, USA **3** School of Biological Sciences, Georgia Institute of Technology, Atlanta, Georgia, USA ¶ Corresponding author * These authors contributed equally.

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Software

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Summary

Tagore is a Python utility for the illustration of genomic features on human chromosome ideograms. It was designed for the visualization of local ancestry tracts, but it can be used for a wide variety of genomic features, including genetic variants and chromatin states. Tagore is freely distributed on several platforms, easy to use, and supports the generation of high-quality, publication-ready images.

Statement of need

A wide variety of functional, comparative, and computational genomics techniques are used to link the DNA sequence of the human genome to biology, health, and evolution (Ernst & Kellis, 2010; Frazer et al., 2004; Harrow et al., 2012; Siepel et al., 2005; Stein, 2001). Application of these techniques yields genome feature annotations, which are coordinate-based delineations and descriptions of the functional and evolutionary elements in the genome. Genome annotations include gene locations along with their exon-intron structure, distributions of epigenetic features, such as methylation sites and histone modifications, and the locations of genetic variants, including single nucleotide variants (SNVs), insertion-deletions (indels), and copy number variants (CNVs). Visualization of genome annotations helps to provide an intuitive sense for their distribution and relationships and is widely used in the scientific literature (Nielsen et al., 2010).

The Tagore utility allows users to illustrate genome annotations on human chromosome ideograms – diagrammatic representations of human chromosomes showing their relative sizes together with the positions of the centromeres and chromosome arms. Tagore was designed to visualize genome-wide distributions of ancestry in the form of local ancestry tracts that illustrate the ancestral origins for specific chromosomal segments (loci) (Figure 1). Nonetheless, the visualization utility provided by Tagore is generic and can be used to support a wide variety of genome feature annotations, including SNVs, indels, CNVs, and epigenetic states.

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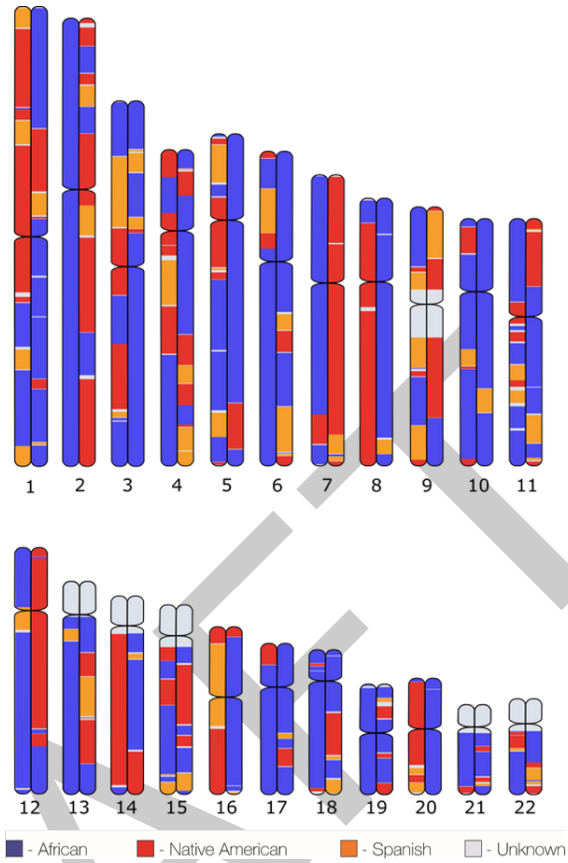

Figure 1: image

Figure 1. **Local ancestry ideogram produced by Tagore.** Chromosomal regions (loci) with African (blue), European (orange), and Native American (red) ancestral origins are shown for 22 human autosomes. Regions with unknown ancestry are shown in gray. This genome is from a participant in the ChocoGen research project (https://www.chocogen.com/) from the Colombian department of Chocó (Conley et al., 2017; Medina-Rivas et al., 2016).

Tagore has been used in several studies to illustrate local ancestry tracts along the human genome, and it has been incorporated into several independently genome annotation utilities (Astore et al., 2023; Medina-Munoz et al., 2023; Norris et al., 2019, 2020; Padilla-Iglesias et al., 2024; Rishishwar et al., 2015). For example, Tagore has been integrated with local ancestry inference packages like Gnomix (https://github.com/AI-sandbox/gnomix) and rf-mix reader (https://rfmix-reader.readthedocs.io/en/latest/index.html) (Benjamin, 2024; Hilmarsson et al., 2021; Maples et al., 2013). It has also been included as part of the CoFrEE method to visualize CNV data (Gaenko et al., 2023). Finally, the need for a utility like Tagore is underscored by the fact that its bioconda package has been downloaded 9,705 times (as of March 14, 2025).

Implementation

Tagore is written in Python and requires Python 3.6+ (https://www.python.org/downloads/), the RSVG libarary for rendering SVG files (https://cran.r-project.org/web/packages/rsvg/index.html) , and the Click package for creating command line interfaces (https://click.palletsprojects.com/en/stable/). It can be installed from GitHub (https://github.com/jordanlab/tagore), PyPI (https://anaconda.org/bioconda/tagore), or bioconda (https://anaconda.org/bioconda/tagore). The Tagore utility merges the genome annotation visualization and image generation process, which would otherwise be slow and labor intensive, entailing a series of manual data

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integration and conversion steps, into a single step that is fast and easy to implement. It takes Browser Extendable Data (BED) format-like input files, where each row represents a genomic feature on a specific chromosome, and generates high resolution, vector graphics images that are publication-ready. Input file rows include chromosome coordinates, feature shapes and sizes, and the chromosome copy (haplotype) on which to render the feature. Tagore supports both hg37 and hg38 human genome builds.

The name Tagore is an homage to Rabindranath Tagore, a Nobel Laureate and major figure of the Bengal Renaissance. Tagore spoke out against racial prejudice and espoused the principle respect for all people, regardless of ancestry or ethnic background.

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